

Name: _____ Student ID: _____

HOMEWORK 3

Fourier Series & Transform

2110203 COMPUTER ENGINEERING MATHEMATICS II

Generative AI Usage Declaration

- ☐ No AI Used: I did not use any AI tools to complete this homework.
- ☐ AI Assisted: I used AI tools as detailed below:

Where	AI model	How

Student Acknowledgement : I declare that I have reviewed, understood, and tested all AI-generated output. I take full responsibility for the accuracy, logic, and functionality of the final submitted work.

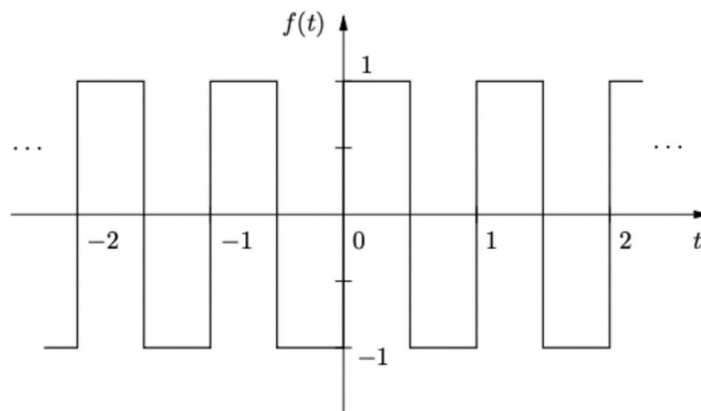
Signature: _____ Date: _____

Name: _____

Student ID: _____

Write solutions and give explanations (1 point)**SEND ONLY PROBLEM 1. and 2.**

1. (0.25 points) Find the Fourier series of the following periodic function



2110203: COMP ENG MATH II (2026/1) Homework 3

Name: _____

Student ID: _____

2. (0.75 points, 0.25 points each) Find the Fourier Series of the periodic function $x(t)$ which are provided as follows. (Hope you are prepared and have review integration by parts)

2.1. $x(t) = \frac{\pi t^3}{2}$; $-1 < t < 1$

Name:

Student ID:

2.2. $x(t) = \pi - t$; $-\pi \leq t \leq \pi$

Name:

Student ID:

2.3. $x(t) = t^2 + \sin^3(\pi t); -1 \leq t \leq 1$

Name: _____

Student ID: _____

OPTIONAL, FOR PRACTICE ONLY

3. Find the Fourier transform of the following signals in terms of $X(j\omega)$, the Fourier transform of $x(t)$
($\mathcal{F}\{x(t)\} = X(j\omega)$)

3.1. $x(-t)$

3.2. $x_e(t) = \frac{x(t)+x(-t)}{2}$

3.3. $x_o(t) = \frac{x(t)-x(-t)}{2}$